

Experimental Psychology 30 - Assignment 3

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Part One: Multiple-Choice Questions (14 marks)

Letter	Statement
The following questions require you to have completed the first three topics in the <i>Central Tendencies Exercise</i> from the LearnAlberta website. If you have not already completed the exercises, please do so before attempting the following multiple-choice questions. Indicate the letter of your choice of each BEST answer.	
<i>Questions 1 to 3 refer to the hockey example.</i>	
C	1. The mean score was 20 goals/5 games = 4 A. 2 B. 3 C. 4 D. 5
A	2. The mode was 2 out of the 5 games had 2 goals therefore most frequent # of goals was 2 A. 2 B. 3 C. 4 D. 5
C	3. The median was A. 2 B. 3 C. 4 D. 5
<i>Questions 4 to 6 refer to the curling example.</i>	

ANSWER KEY

A	4. The mean was A. 7.0 B. 7.1 C. 7.2 D. 7.3
A	5. The mode was A. 6 and 9 B. 6 C. 9 D. 7.5
C	6. The median was A. 7.0 B. 7.1 C. 7.2 D. 7.3
Questions 7 to 9 refer to the Math 14 example with five (5) tests.	
A	7. The mean was A. 68.0 B. 68.1 C. 68.2 D. 68.3
A	8. The mode was A. 68.0 B. 68.1 C. 68.2 D. 68.3

ANSWER KEY

C	9. The median was A. 68.0 B. 68.1 C. 68.2 D. 68.3
<i>Questions 10 to 12 refer to the Math 14 example with six (6) tests.</i>	
C	10. The mean was $441/6$ A. 73.3 B. 73.4 C. 73.5 D. 73.6
A	11. The mode was A. 68 B. 69 C. 70 D. 71
B	12. The median was A. 68 B. 69 C. 70 D. 71
C	13. When a researcher calculates a large standard deviation, the numbers in the data set are A. very small values B. clumped closely together C. spread far apart D. very large values
A	14. Which of the following is a measure of variability? A. Range B. Mean C. Mode D. Distribution

Part Two: Written-Response Questions (12 marks)

1. Is it possible to score in the 65th percentile yet not pass an exam? Explain. **(2 marks)**

A percentile is not the same as a percentage. If Lance scored 32 of a possible 50 marks on a test for introversion, his percentage score would be 64%. To determine the percentage, you divide 32 by 50 then multiply the quotient by 100. If Lance wanted to compare his score to the scores of others on the same test, he would use his percentile. If 70 percent of the other scores were less than Lance's score (less than 32 out of 50) then Lance's score would be the 70th percentile. To summarize, percentiles compare your score to the scores of others who have taken the same test.

2. Define reliability as it pertains to scientific experimentation and state why reliability is important. **(4 marks)**

Reliability refers to the results of an experiment being obtained that are similar when the experiment is repeated by others. The three types of reliability are test-retest, internal, and inter-rater. The conclusions reached are considered reliable if they can be transferred to similar situations in other parts of the world.

3. Define negative feedback and provide an example of a system that uses negative feedback. Do not repeat an example provided in the course content. **(4 marks)**

Negative feedback tends to reduce the future output of a system. Examples should not include a home heating system, personal heating system (body), or human hormone systems.

4. Can the results of an experiment be ascribed to groups other than those that participated in the experiment? Explain. **(2 marks)**

No - Researchers discovered that the results from studies that only focus on one subset of individuals could not simply be transferred over to other demographic groups. How women, children, and members of other races react to drugs, therapies, suggestions, and other treatment methodologies is often different from how a Caucasian adult male reacts so you could not transfer results from Caucasian adult males to other demographic groups..

Part Three: Measures of Central Tendency and Measures of Variability (8 marks)

Select either **Option A** or **Option B**. Use the table below to answer the corresponding set of questions. Please show your calculations.

Table: Hand width of 10 grade four students.

7 cm	11 cm	10 cm	8 cm	8 cm	7 cm	9 cm	12 cm	10 cm	8 cm
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Option A: (Easier)

1. To what do the terms of central tendency (mode, median, and mean) refer?

Central tendency refers to how clustered a group of numbers is. Mathematically, central tendency refers to the average in a set of data.

2. What is the mode?

The mode is the number that occurs most often in a set of data. To determine the mode, you should first organize the numbers in order of size. 7 7 8 8 8 9 10 10 11 12
Mode = 8

3. What is the median?

The median is the number in the middle of the set of data after you arrange the numbers in order of size. In many cases, you will have the same number of values on the left and right of the median value. If, however, there is an even number of data points, you will have two numbers as your median and the values must be averaged.
7 7 8 8 8 9 10 10 11 12 median: $8 + 9 = 17$, $17 \div 2 = 8.5$

4. What is the mean?

The mean is the value obtained when you add all the values in a data set and then divide the sum (total) by the number of values in the data set.
Mean: $7 + 7 + 8 + 8 + 8 + 9 + 10 + 10 + 11 + 12 = 90$, $90 \div 10 = 9$

Option B: (More Difficult)

ANSWER KEY

1. To what do the terms of variability (range, variance, standard deviation) refer?

The measures of variability refer to how spread out a set of numbers is. The terms variability, distribution, and dispersion are synonyms and refer to the spread of a data set.

2. What is the range?

The least complicated measure of variability is the range. The range is the difference between the highest value in a data set and the lowest value in a data set.
Range: $12 - 7 = 5$

3. What is the variance?

Using the mean as the measure of the middle of a distribution, researchers calculate the average squared difference of the scores from the mean. The result of this calculation is called the variance.
Determine the arithmetic average or mean from the data set by adding all n values and dividing the sum by n. $\text{mean} = 9$
.Calculate the difference between each score and the mean value, and then square that difference. $7 - 9 = -2$; $8 - 9 = -1$; $9 - 9 = 0$; $10 - 9 = 1$; $11 - 9 = 2$; $12 - 9 = 3$
 $2^2 = 4$; $1^2 = 1$; $0^2 = 0$; $1^2 = 1$; $2^2 = 4$; $3^2 = 9$
Determine the arithmetic average or mean of all the squared differences by adding all six calculated values together and dividing the sum by six. $4 + 1 + 0 + 1 + 4 + 9 = 19$
 $19 \div 6 = 3.2$

4. What is the standard deviation?

Determine the standard deviation by finding the square root of the variance.
If the value of standard deviation is small, the numbers in a data set are clustered closely together. If a standard deviation is large, then the numbers are more spread out. $\text{s.d.} = \sqrt{3.2} = 1.79$

End of Assignment 3

When you have completed your assignment, upload this document to the Assignment 3 drop box on Brightspace.